general scattering equation

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The time-independent Schrödinger equation is (atomic units are used):

$$\left(-\frac{1}{2}\nabla^2 + V(\mathbf{r})\right)\Psi_j(\mathbf{r}) = E\,\Psi_j(\mathbf{r}). \tag{1}$$

where 'j' is a linearly independent solution.

$$V(\mathbf{r})\Psi_j(\mathbf{r}) = V_{st}(\mathbf{r})\Psi_j(\mathbf{r}) - \left(V_{exc}\Psi_j\right)(\mathbf{r}), \tag{2}$$

so that

$$\left(\nabla^2 + 2E - 2V_{st}(\mathbf{r})\right)\Psi_j(\mathbf{r}) = -2\left(V_{exc}\Psi_j\right)(\mathbf{r}). \tag{3}$$

$$\left(V_{exc}\Psi_j\right)(\mathbf{r}) = \alpha \sum_{i\in occ} \Phi^i(\mathbf{r})\, F^i_j(\mathbf{r}), \tag{4}$$

$$F^i_j(\mathbf{r}) = \frac{1}{\alpha}\int d^3r'\, \frac{\Phi^{i*}(\mathbf{r'})\,\Psi_j(\mathbf{r'})}{|\mathbf{r}-\mathbf{r'}|}. \tag{5}$$

where $\alpha = \sqrt{2\pi}$ is chosen somehow that the superpotential matrix becomes symmetric.

$$\nabla^2 F^i_j(\mathbf{r}) = -\frac{4\pi}{\alpha}\,\Phi^{i*}(\mathbf{r})\,\Psi_j(\mathbf{r}). \tag{6}$$

$$\Psi_j(\mathbf{r}) = \sum_\mu \frac{\psi_{\mu j}(r)}{r}\, Y_\mu(\hat{\mathbf{r}}), \tag{7}$$

$$F^i_j(\mathbf{r}) = \sum_\sigma \frac{f^i_{\sigma j}(r)}{r}\, Y_\sigma(\hat{\mathbf{r}}), \tag{8}$$

$$\Phi^i(\mathbf{r}) = \sum_\lambda \frac{\chi^i_\lambda(r)}{r}\, Y_\lambda(\hat{\mathbf{r}}), \tag{9}$$

with orthonormal spherical harmonics

$$\int d\Omega\, Y^*_\mu(\Omega) Y_{\mu'}(\Omega) = \delta_{\mu\mu'}. \tag{10}$$

$$\nabla^2\left(\frac{u(r)}{r} Y_{\ell m}(\hat{\mathbf{r}})\right) = \frac{1}{r}\left(u''(r) - \frac{\ell(\ell+1)}{r^2}u(r)\right) Y_{\ell m}(\hat{\mathbf{r}}). \tag{11}$$

$$C_{\sigma\lambda\mu} = \int d\Omega\; Y^*_\sigma(\Omega)\, Y^*_\lambda(\Omega)\, Y_\mu(\Omega). \tag{12}$$

$$f^{i\,\prime\prime}_{\sigma j}(r) - \frac{\ell_\sigma(\ell_\sigma+1)}{r^2}\, f^i_{\sigma j}(r) = -\frac{4\pi}{\alpha}\sum_{\lambda,\mu} C_{\sigma\lambda\mu}\, \frac{\chi^{i*}_\lambda(r)\,\psi_{\mu j}(r)}{r}. \tag{13}$$

$$V^{st}_{\mu\mu'}(r) = \int d\Omega \, Y^*_\mu(\Omega) \, V_{st}(r,\Omega) \, Y_{\mu'}(\Omega). \tag{14}$$

$$G_{\mu\lambda\sigma} = \int d\Omega \, Y^*_\mu(\Omega) \, Y_\lambda(\Omega) \, Y_\sigma(\Omega). \tag{15}$$

$$\psi''_{\mu j}(r) - \frac{\ell_\mu(\ell_\mu+1)}{r^2}\psi_{\mu j}(r) + 2E\,\psi_{\mu j}(r) - 2\sum_{\mu'} V^{st}_{\mu\mu'}(r)\,\psi_{\mu' j}(r) = -2\alpha \sum_{i\in occ}\sum_{\lambda,\sigma} G_{\mu\lambda\sigma}\,\frac{\chi^i_\lambda(r)\, f^i_{\sigma j}(r)}{r}. \tag{16}$$

$$V_{\mu\mu'}(r) = \frac{\ell_\mu(\ell_\mu+1)}{2r^2}\delta_{\mu\mu'} + V^{st}_{\mu\mu'}(r), \tag{17}$$

$$Q^i_{\mu\sigma}(r) = \sum_\lambda G_{\mu\lambda\sigma}\,\frac{\chi^i_\lambda(r)}{r}, \tag{18}$$

so that

$$\sum_{\mu'}\left[\psi''_{\mu j}(r)\,\delta_{\mu\mu'} + 2\left(E\,\delta_{\mu\mu'} - V_{\mu\mu'}(r)\right)\psi_{\mu' j}(r)\right] = -2\alpha\sum_{i\in occ}\sum_\sigma Q^i_{\mu\sigma}(r)\, f^i_{\sigma j}(r). \tag{19}$$

$$k = \sqrt{2E}. \tag{20}$$

$$\lim_{r\to 0}\psi_{\mu j}(r) \sim \hat{j}_{\ell_\mu}(kr), \tag{21}$$

$$\lim_{r\to\infty}\psi_{\mu j}(r) \sim A_{\mu j}\,\hat{j}_{\ell_\mu}(kr) + B_{\mu j}\,\hat{y}_{\ell_\mu}(kr). \tag{22}$$

$$\lim_{r\to 0} f^i_{\sigma j}(r) \sim r^{\ell_\sigma+1}, \tag{23}$$

$$\lim_{r\to\infty} f^i_{\sigma j}(r) \sim r^{-\ell_\sigma}. \tag{24}$$

$$S = (I+iK)(I-iK)^{-1}. \tag{25}$$